

Beyond Newton's Golden Formula: Extending Sir Isaac's physics to unify with Albert Einstein's general relativity.

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Category: Relativity and Cosmology / Gravity

By considering the universe as a fluid and Gravity as an electromagnetic monopole field where gravity only attracts, even to itself. And applying the proper fluid viscosity for space. The below formula can be used to calculate SMB's, Galactic orbits, and Mercury's orbital wobble without tensors derivatives or matrices. This formula applied in this manner behaves the same way tensors behave by providing additional energy in high gravity.

$$F = \frac{GMm}{r^2} + \frac{GMm \cdot a}{r^3}$$

1. **F** = Total Gravitational Force
2. **G** = The Gravitational Constant
3. **M** = Mass of Body A
4. **m** = Mass of Body B
5. **r** = Distance between them
6. **a** = The Field Interaction overlap (the actual amount of the gravitational force that is acting on itself and other gravitational fields individually and as a combinatory effect at once derived from the object's mass and the inverse square law)
 - 6.1. Clarification on the Physical Nature of Variable a
 - 6.2. Note on Real-World Application vs. Computational Modeling:
 - 6.3. In nature, the field interaction variable a is not a static scalar, nor is it a simple point-to-point coordinate calculation. Because space behaves as a continuous fluid medium, a gravitational field is a localized pressure gradient that physically stacks, overlaps combinatorially.
 - 6.4. 1. An object's independent field profile $1/r^2$
 - 6.5. 2. The immediate local interaction or the gravity interaction with gravity $1/r^3$
 - 6.6. 3. The gravitational field interaction of the local object and the galactic gravitational field $1/r^3$

All objects have to be calculated locally, and with their gravitational interaction with other gravitational fields, and the interaction with the gravitational force of the galaxy and its field interactions. And all sides of their gravitational field must be considered and gravity always pulls.

Environmental Setup.

1. Space considered the fluid medium
 - 1.1. Set fluid density to space
 - 1.2. Decouple fluid density and resistance
 - 1.3. Resistance added via viscosity variable
 - 1.4. Time is set as a localized step rate relative to the gravitational force concentration
 - 1.4.1. This mimics time dilation
2. Gravity is set as an electromagnetic monopole field with specific rules
 - 2.1. The gravitational energy count is set 2=1
 - 2.1.1. Minimum floor ½ of “1 whole” (2=1)
 - 2.1.1.1. Akin to the Planck limit, not a calculation limit.
 - 2.1.2. Naturally creates the inverse square law
 - 2.1.3. Prevents from dividing by zero
 - 2.2. Gravity always attracts (even to itself)
 - 2.2.1. Gravitational force $1/r^2$
 - 2.2.2. Gravitational self attraction $1/r^3$
 - 2.2.3. Prevents shockwaves wake effects
 - 2.3. Other variables like mass, speed, rotation, heat, spin, and any other factors relevant to the environment are added as interference variables and concentration patterns
3. Utilize proven fluid dynamics and electromagnetic monopole models and formulas

I have modeled this math in 3 different python programs. Orbit.py simulating a Mercury like planet orbiting near a sun, Gal.py A Galaxy model, and SMB.py a black hole model. My primary formula seamlessly calculates mercury's wobble, predicts the accelerated speed of objects at the edge of galaxies, and mapped SMB rotations correctly without loss or crash. All as models matched real world observed results without the need to introduce dark matter or dark energy. The SMB model is not calculating the individual field of the planets with the interactive force simultaneously and cannot parse the gravity gradient and calculates from center mass not individual center and overall interactions because of the way my processor applies and the order it calculates not because of the math or the actual formula.

$$F = \frac{GMm}{r^2} + \frac{GMm \cdot a}{r^3}$$

This added property of gravity itself is all that is needed to completely remove the need for tensors.

Programs and results can be found here:

Orbit.py <https://github.com/Stephenscience/Beyond-Golden/blob/main/Orbit.py>
orbit output.pdf <https://github.com/Stephenscience/Beyond-Golden/blob/main/orbit%20output.pdf>
SMB.py <https://github.com/Stephenscience/Beyond-Golden/blob/main/SMB.py>
SMB results.pdf <https://github.com/Stephenscience/Beyond-Golden/blob/main/SMB%20results.pdf>
Gal.py <https://github.com/Stephenscience/Beyond-Golden/blob/main/Gal.py>
Gal output.pdf <https://github.com/Stephenscience/Beyond-Golden/blob/main/Gal%20output.pdf>